

GATE QUESTION Q36

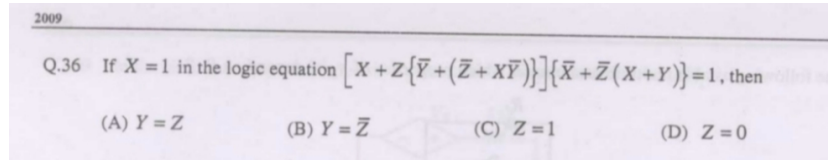


Figure 1: GATE Question Q36

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1 Components

1.1 Breadboard

A breadboard is used to make temporary circuit connections.

1.2 Seven Segment Display

A common anode seven segment display is used to display the output.

1.3 Arduino Uno

Arduino Uno is used to implement the Boolean expression.

1.4 Jumper Wires

Jumper wires are used to connect the circuit components.

2 Boolean Simplification

For (X=1),

$$F = \bar{Z}$$

For output 1,
[Z=0]

3 Hardware Connections

Arduino Pin	Segment heightD2
a D3	b D4
c D5	d D6
e D7	f D8
g height	

The common anode pin is connected to +5V.

4 Software Implementation

```
int Z = 0;  
int F = !Z;
```

```

if(F == 1)
{
display1();
}
else
{
display0();
}

```

5 Output

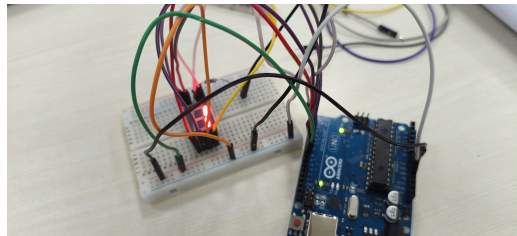


Figure 2: Output showing 1

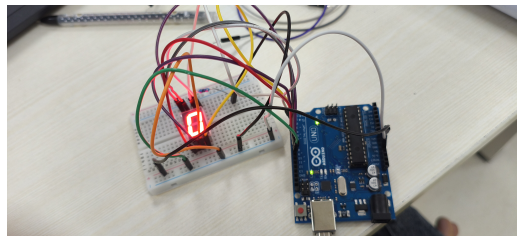


Figure 3: Output showing 0

6 Result

The Boolean expression was implemented using Arduino Uno and a seven segment display.

[F=]

Hence,

[Z=0]

for output 1.